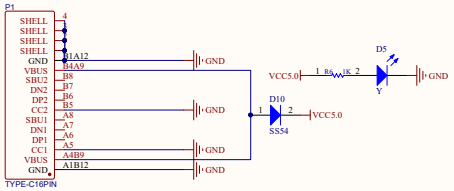
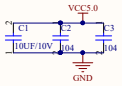
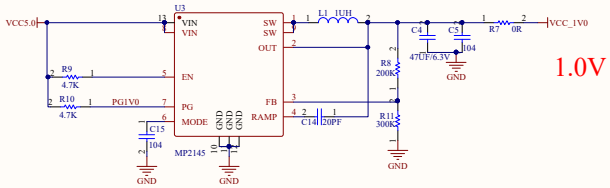
第一部分 ZYNQ的电源系统

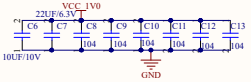
上电顺序：1.0V->1.8V->1.5V->3.3V





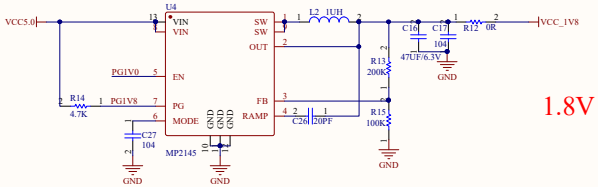


1.0V电源

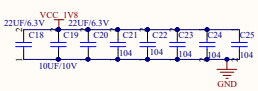


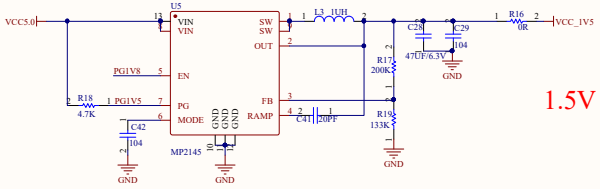
ZYNQ内核电源：1.0V

DDR3的电源：1.5V

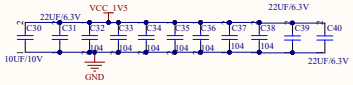


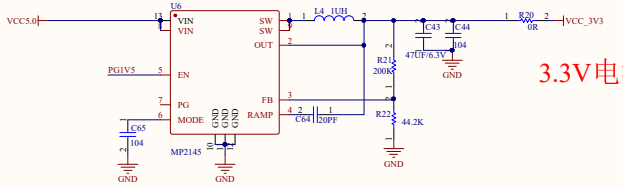
1.8V电源



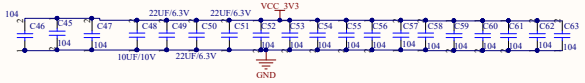


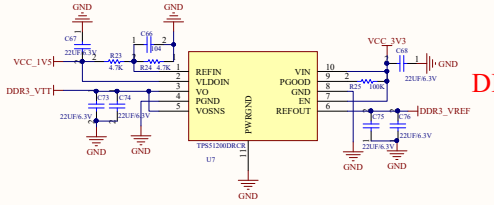
1.5V电源



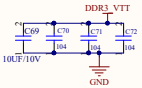


3.3V电源

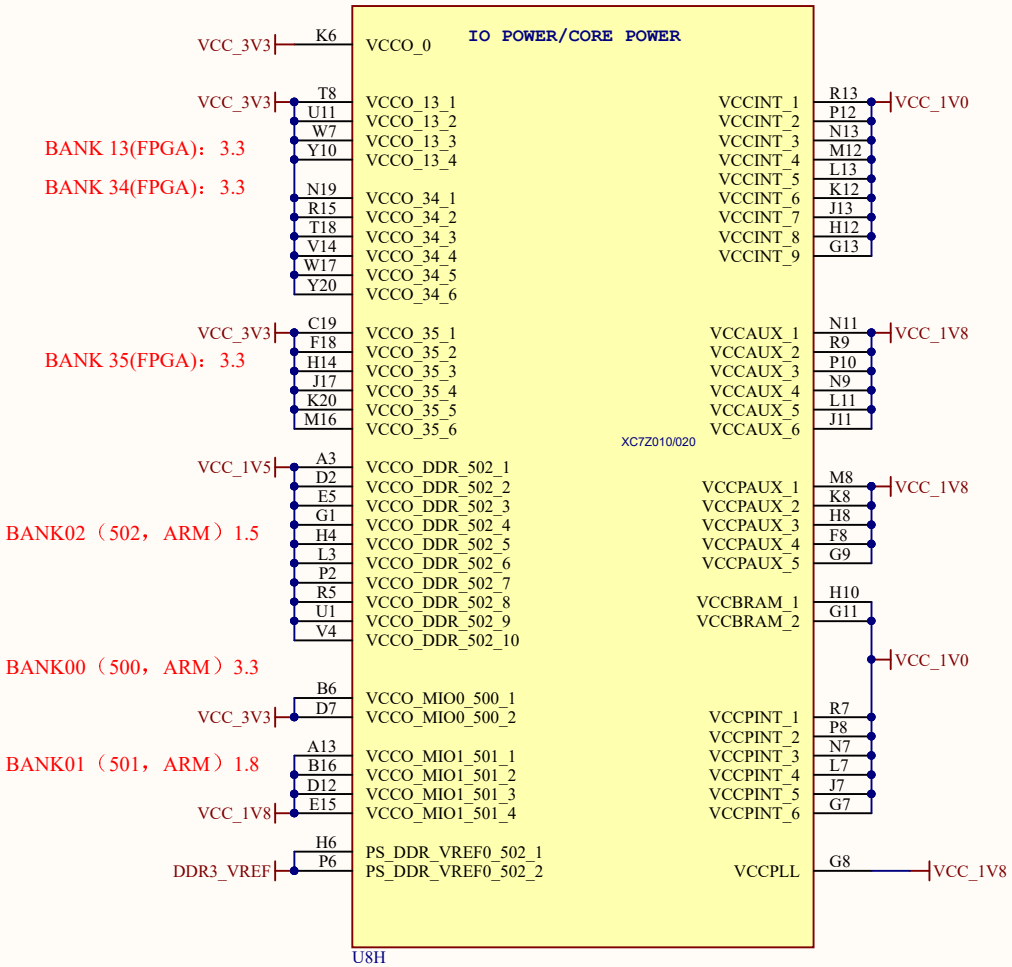


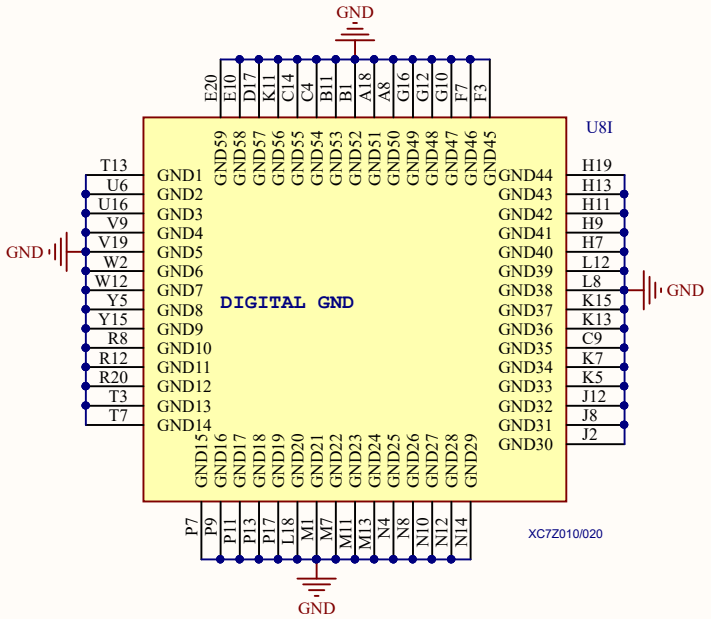


DDR3的VTT和VREF电源



第二部分 ZYNQ芯片的电源引脚

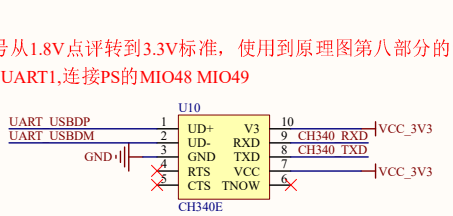


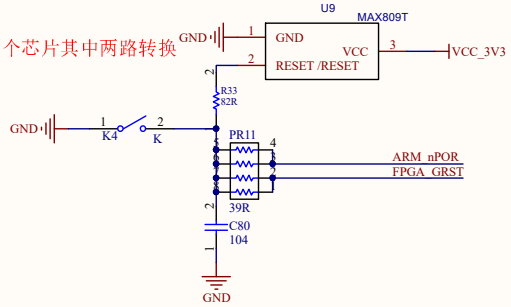


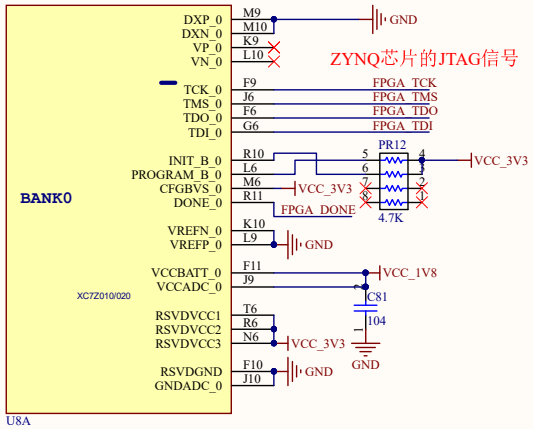
第三部分 ZYNQ的JTAG和调试器电路

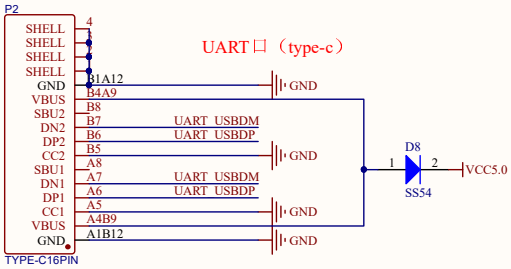
上电复位芯片，检测到3.3电源电压就绪产生一个拉低复位

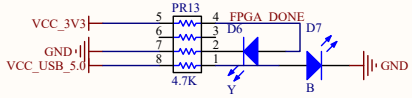
串口信号从1.8V点评转到3.3V标准，使用到原理图第八部分的SD卡电平转换同一个芯片其中两路转换
ZYNQ的UART1,连接PS的MIO48 MIO49

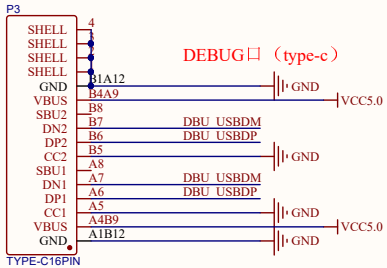


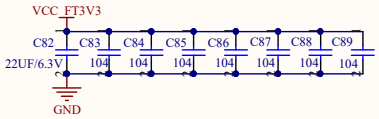


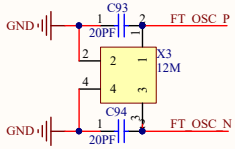


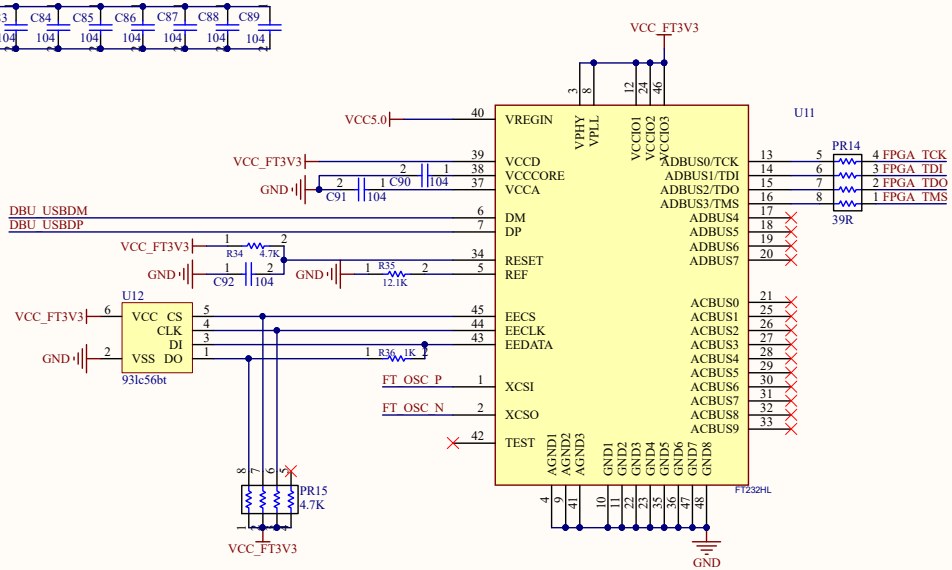




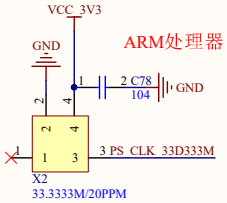




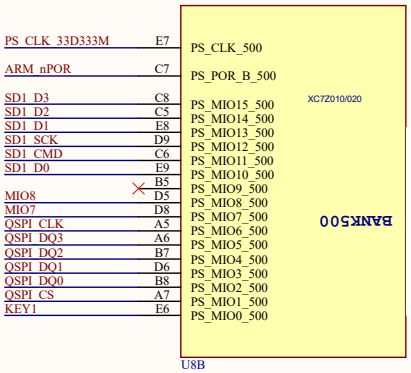




第四部分 ZYNQ的PS部分BANK（仅用于ARM）



ARM的bank0（500）,这个bank所有IO电平3.3V



几个常用的配置MIO[5:2]:

1、JTAG独立配置0001，比较少用；

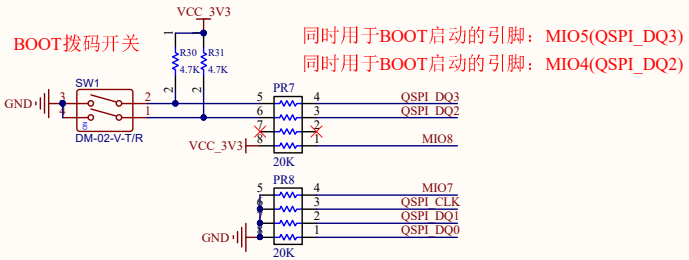
2、JTAG级联配置0000，JTAG调试常用；

3、FLASH启动：Quad-SPI mode，1000；

4、SD card：1100.

平时MIO2、3连接QSPI的存储器，状态上电默认为0

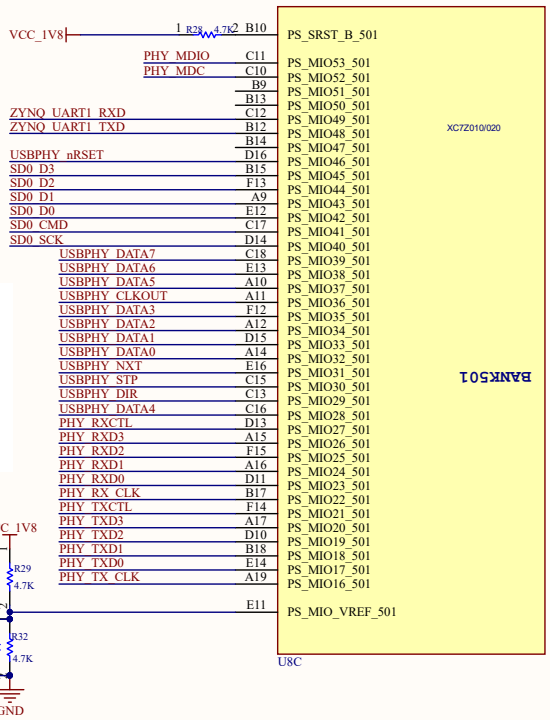
所以不用管,只管MIO4和5，两个决定是FLASH还是SD启动



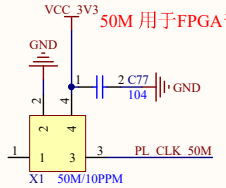
同时用于BOOT启动的引脚：MIO5(QSPI_DQ3)

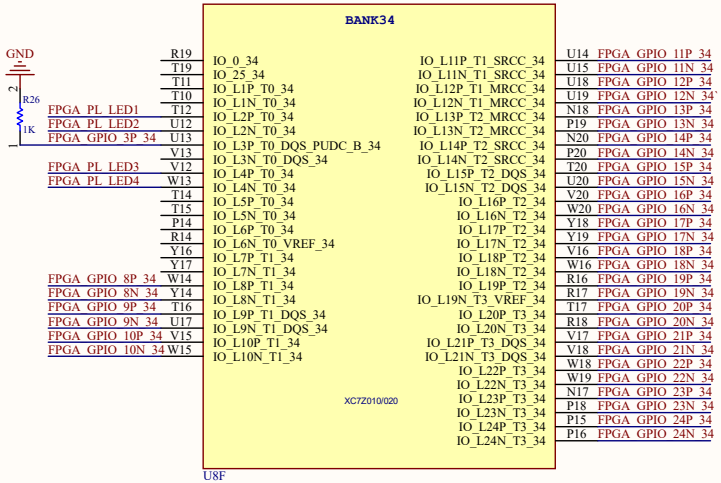
同时用于BOOT启动的引脚：MIO4(QSPI_DQ2)

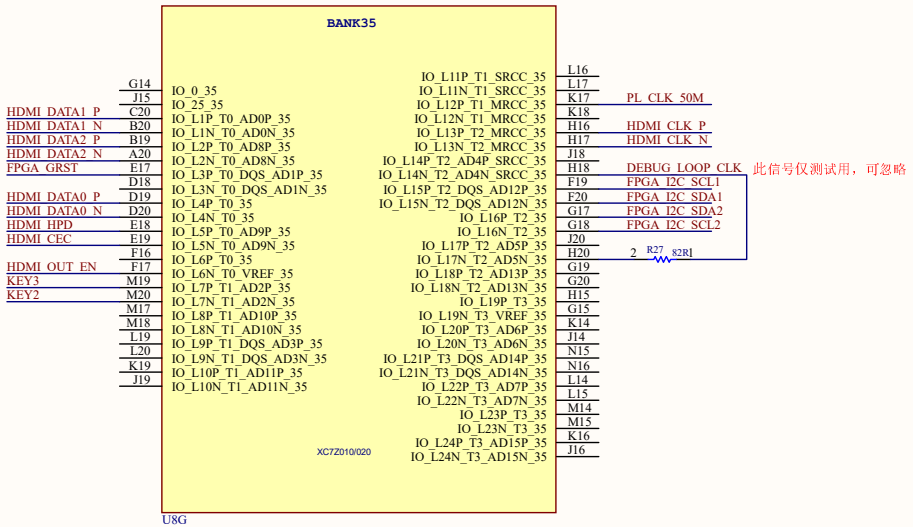
ARM的bank1（501）,这个bank所有IO电平1.8V



第五部分 ZYNQ的PL部分的BANK（FPGA部分）



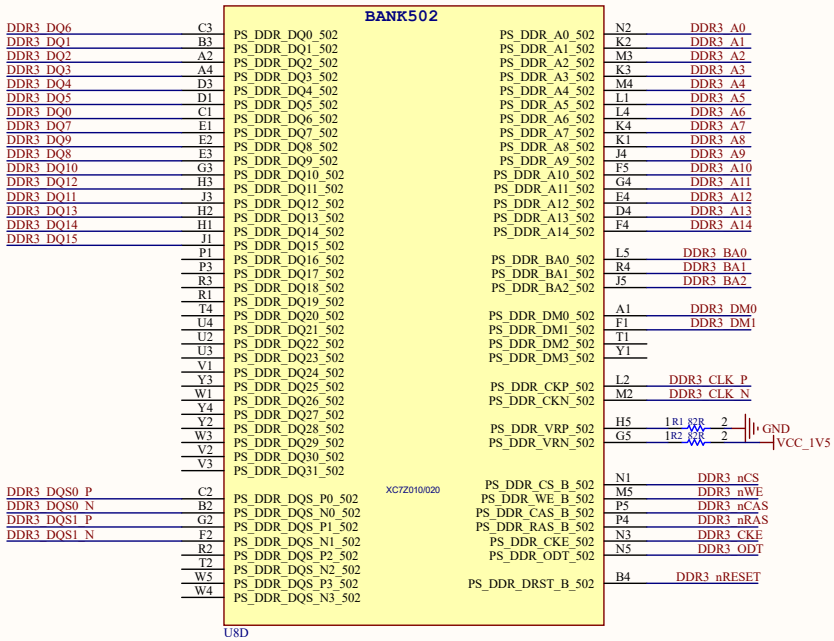




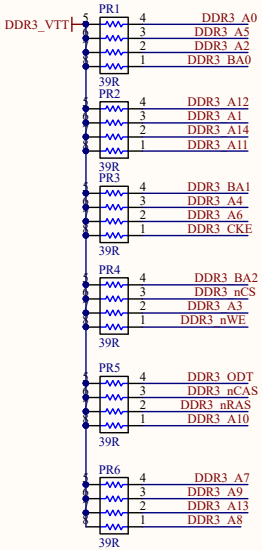
注意：ZYNQ 7020还包含BANK13，但是我们本开发板没有用到BANK13

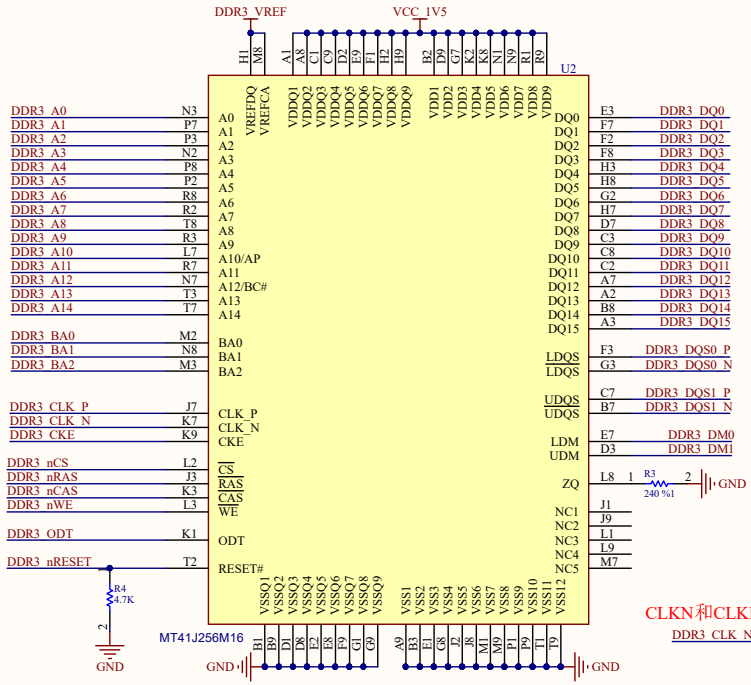
第六部分 ZYNQ的PS部分的DDR3

DDR3的DQ和DQS，DM等长，DQSP和DQSN还需要做差分

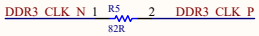


A0-A14以及其控制线需要做等长，并且终端需要匹配39R电阻上拉到VTT

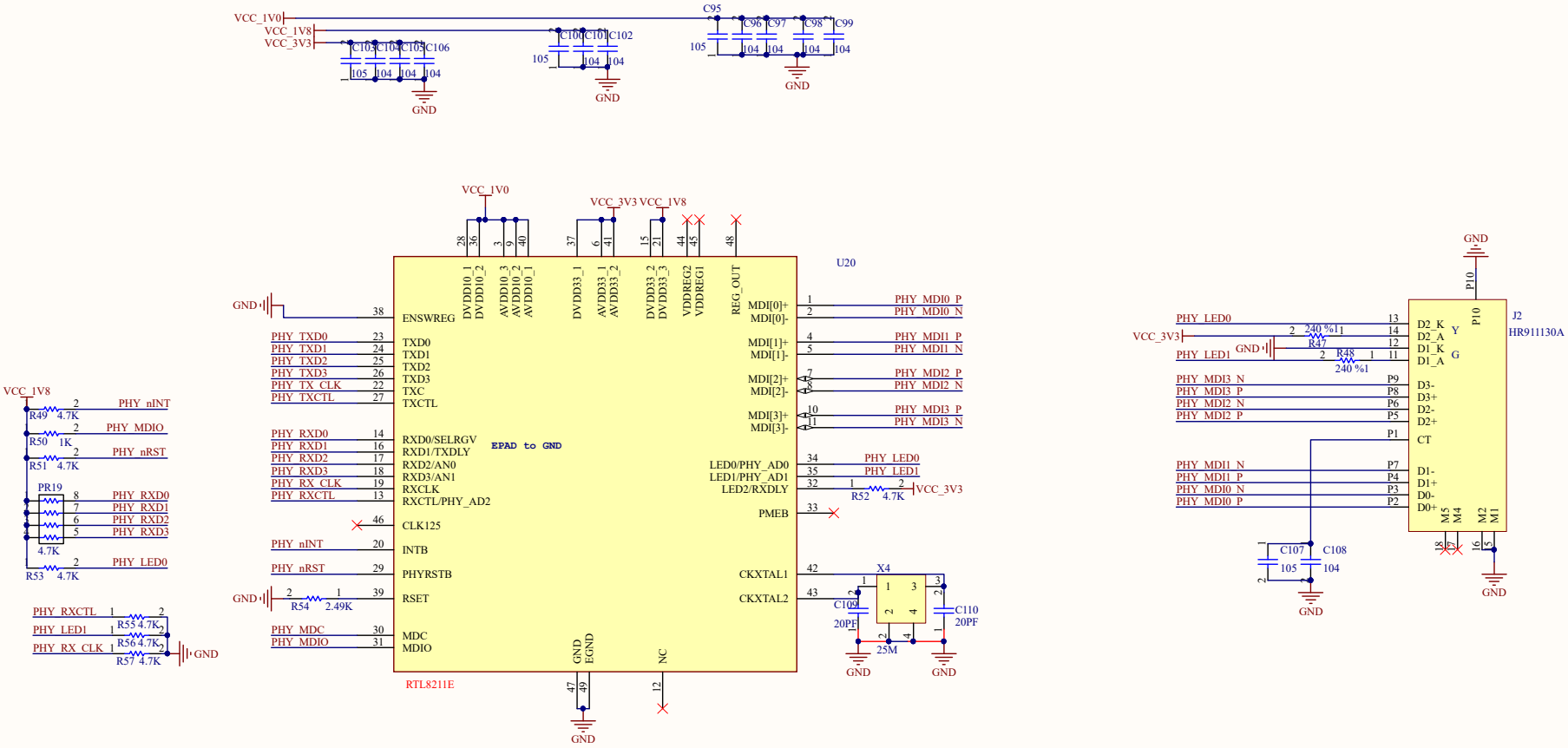




CLKN和CLKP需要做差分，并且与地址线等长

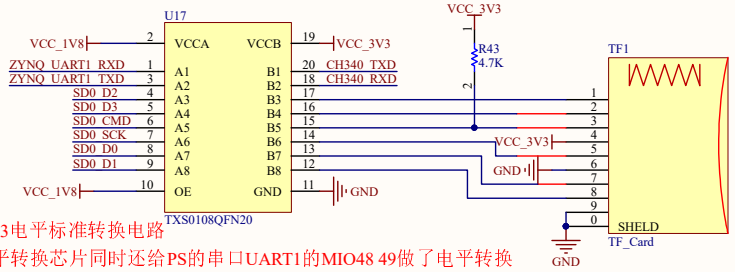


第七部分 ZYNQ的PS部分千兆网口RTL8211E电路

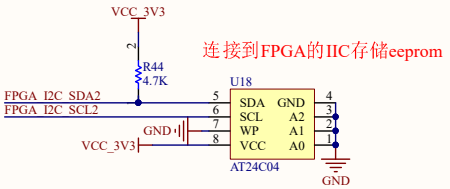


第八部分 ZYNQ存储，QSPI FLASH,SD卡， EEPROM

SD卡座（master，TF1，支持从这个TF卡启动）
SD0
从TF（master）上电启动：BOO设置为11

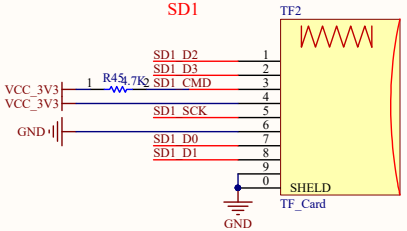


1.8V和3.3电平标准转换电路
这个电平转换芯片同时还给PS的串口UART1的MIO48 49做了电平转换

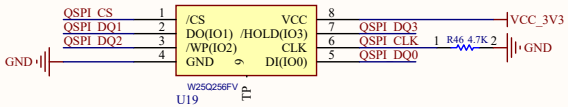


连接到FPGA的IIC存储eeprom

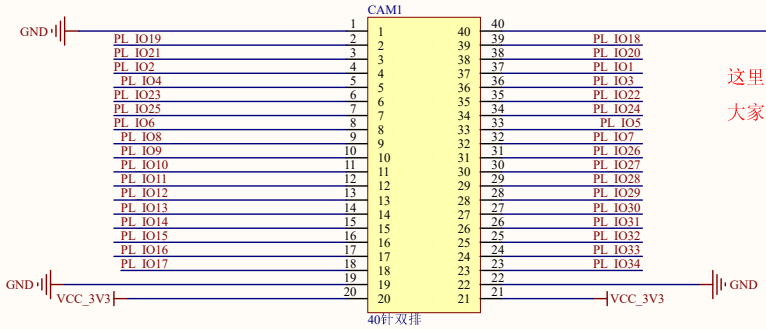
SD卡座（TF2，不支持启动）
SD1



系统的QSPI存储，上电启动
从QSPI存储器启动，BOOT设置为10



第九部分 开发板对外的用户扩展34个IO口及LED和按钮

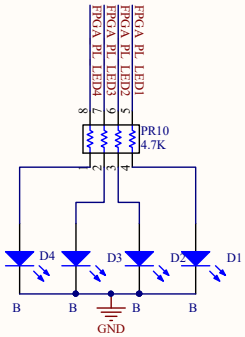


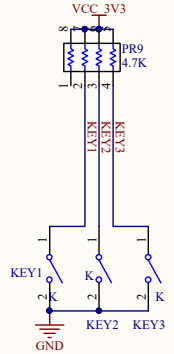
这里的IO排针大家对应原理图的网络标号关系来看

大家可以直接看PCB板子背面的丝印，直接对应ZYNQ芯片的引脚名字



四个LED都是PL的





KEY1:PS的MIO0

KEY2:PL的M20引脚

KEY3:PL的M19引脚

KEY1是PS的，KEY2 KEY3是PL的

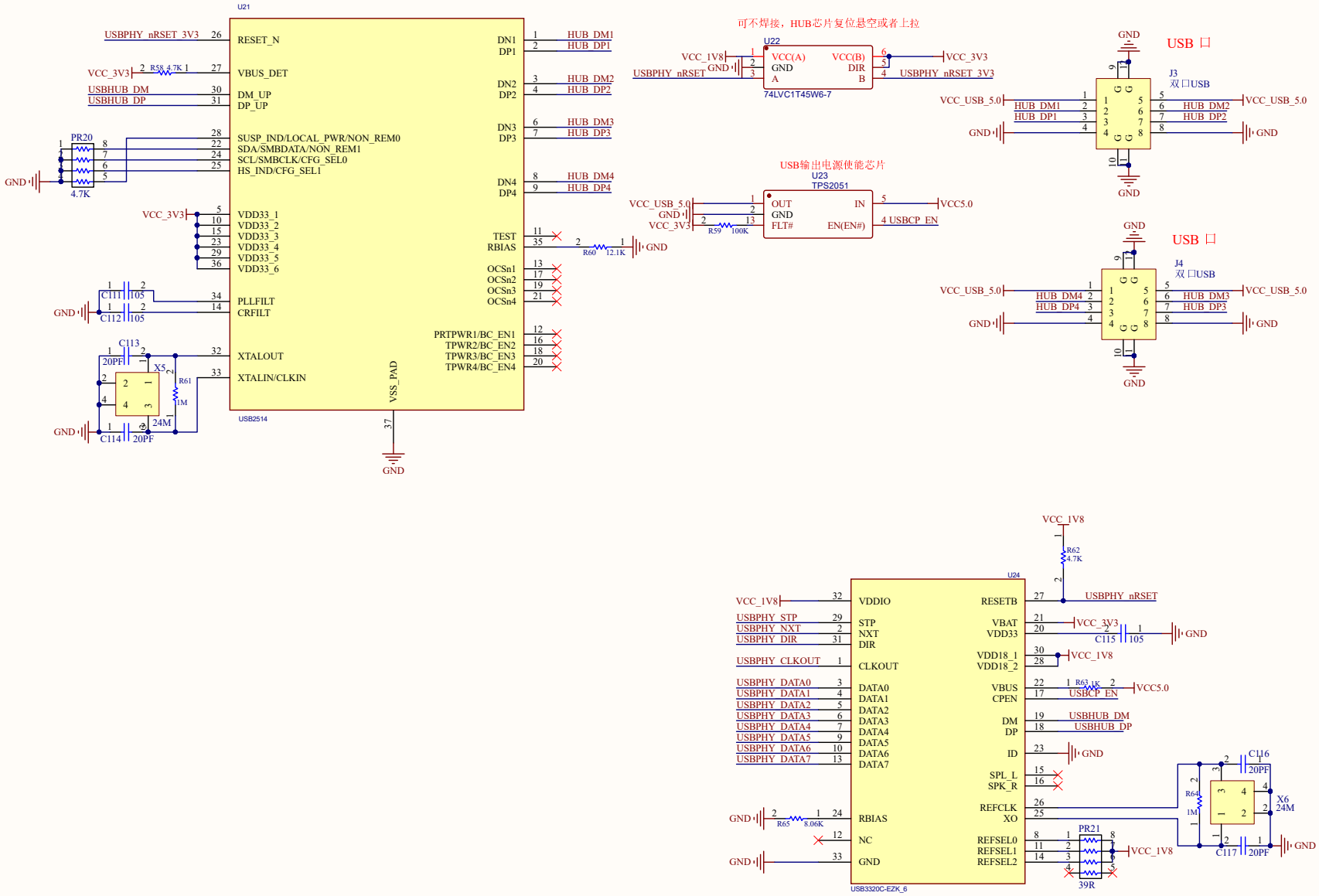
这里只是重新把扩展IO的信号命名一下

具体大家使用IO的时候按照板子对应的丝印标注来约束，电平标准3.3V

FPGA_GPIO_21P_34	PL_IO1
FPGA_GPIO_9N_34	PL_IO2
FPGA_GPIO_17P_34	PL_IO3
FPGA_GPIO_21N_34	PL_IO4
FPGA_GPIO_23P_34	PL_IO5
FPGA_GPIO_11P_34	PL_IO6
FPGA_GPIO_8P_34	PL_IO7
FPGA_GPIO_8N_34	PL_IO8
FPGA_GPIO_18P_34	PL_IO9
FPGA_GPIO_12P_34	PL_IO10
FPGA_GPIO_20P_34	PL_IO11
FPGA_GPIO_19N_34	PL_IO12
FPGA_GPIO_16N_34	PL_IO13
FPGA_GPIO_16P_34	PL_IO14
FPGA_GPIO_15N_34	PL_IO15
FPGA_GPIO_15P_34	PL_IO16
FPGA_GPIO_14N_34	PL_IO17

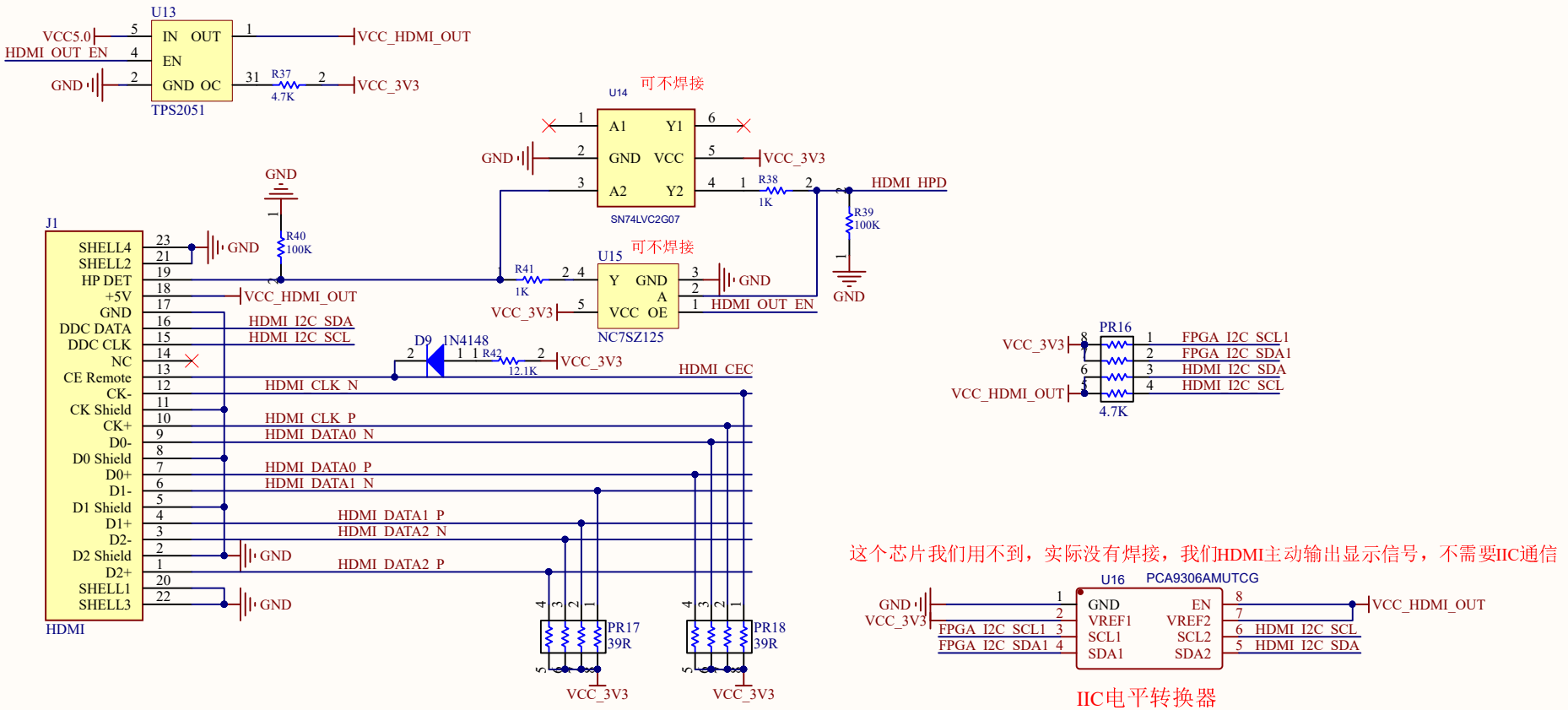
PL_IO18	FPGA_GPIO_24P_34
PL_IO19	FPGA_GPIO_11N_34
PL_IO20	FPGA_GPIO_10P_34
PL_IO21	FPGA_GPIO_10N_34
PL_IO22	FPGA_GPIO_17N_34
PL_IO23	FPGA_GPIO_22P_34
PL_IO24	FPGA_GPIO_22N_34
PL_IO25	FPGA_GPIO_12N_34
PL_IO26	FPGA_GPIO_24N_34
PL_IO27	FPGA_GPIO_19P_34
PL_IO28	FPGA_GPIO_9P_34
PL_IO29	FPGA_GPIO_18N_34
PL_IO30	FPGA_GPIO_20N_34
PL_IO31	FPGA_GPIO_13N_34
PL_IO32	FPGA_GPIO_23N_34
PL_IO33	FPGA_GPIO_13P_34
PL_IO34	FPGA_GPIO_14P_34

第十部分 开发板的USB电路，以及USB HUB电路



注：USB使用PS的USB0,通过HUB芯片一拖四，仅支持HOST

第十一部分 HDMI电路



注：开发板HDMI仅支持输出